$\frac{S}{d^2}$ $\qquad$ $\frac{S}{D^2}$

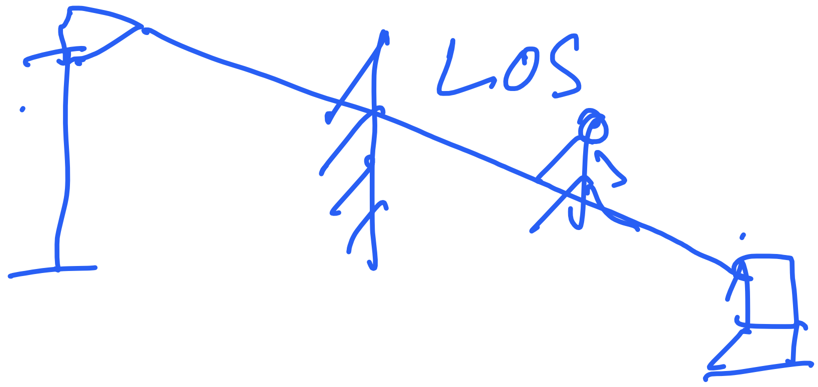
Path-loss $\quad \frac{1}{d^2}$

Tx ——— Rx

↓ $\qquad$ ↓

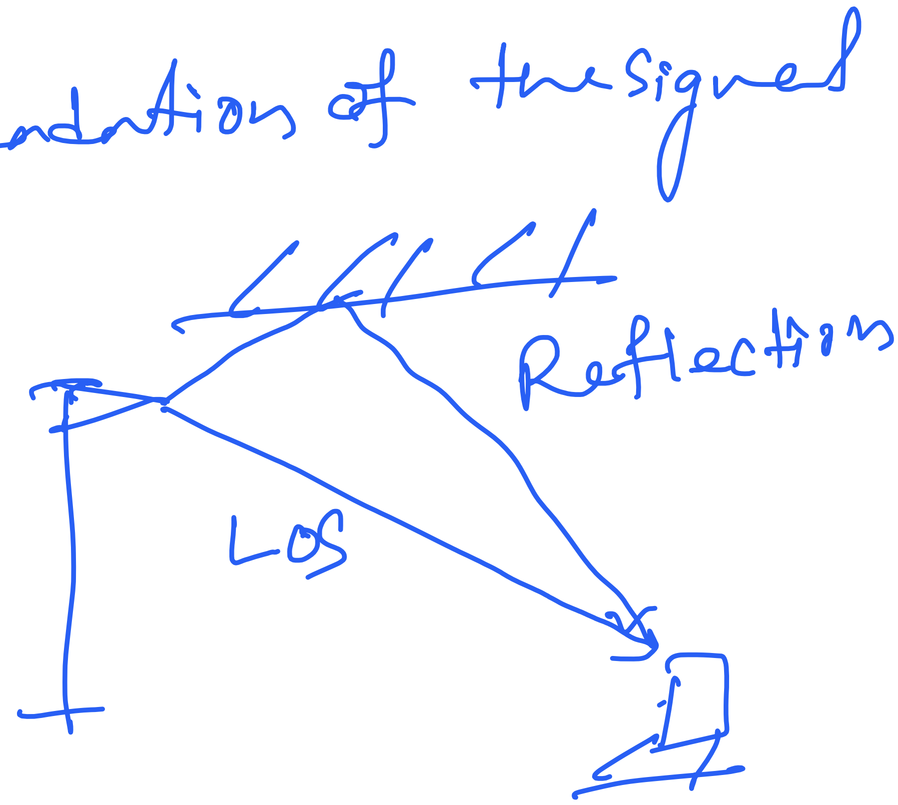
Transmitter $\qquad$ Receiver

LOS → Line of Sight

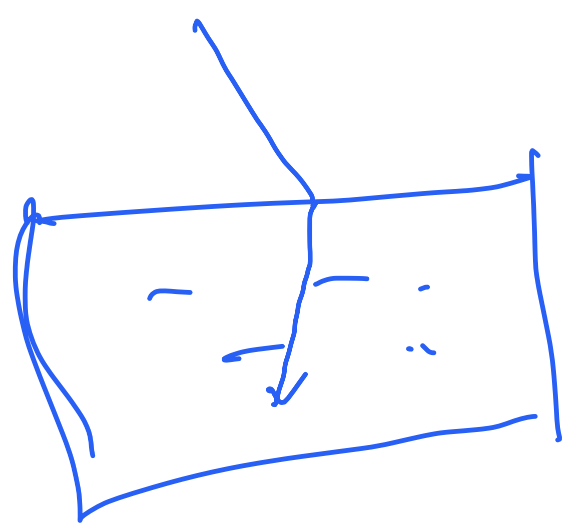


Attenuation → degradation of the signal

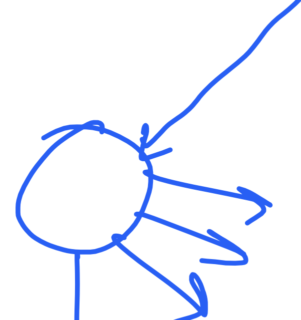
→ reflection

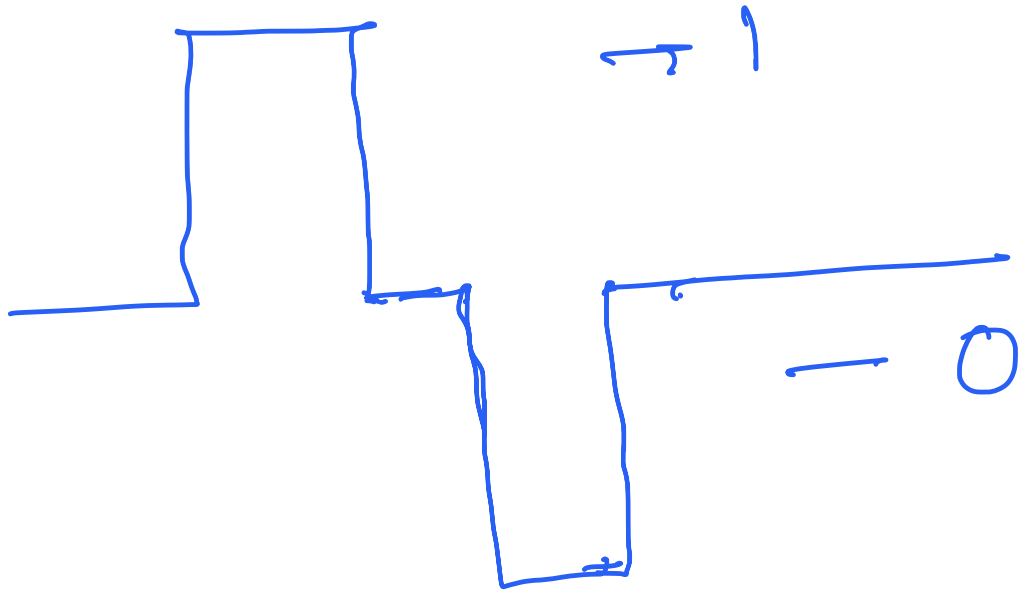


Refraction

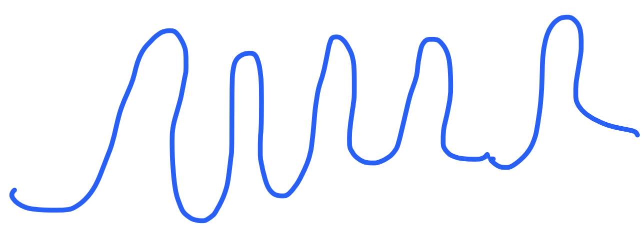


Scattering

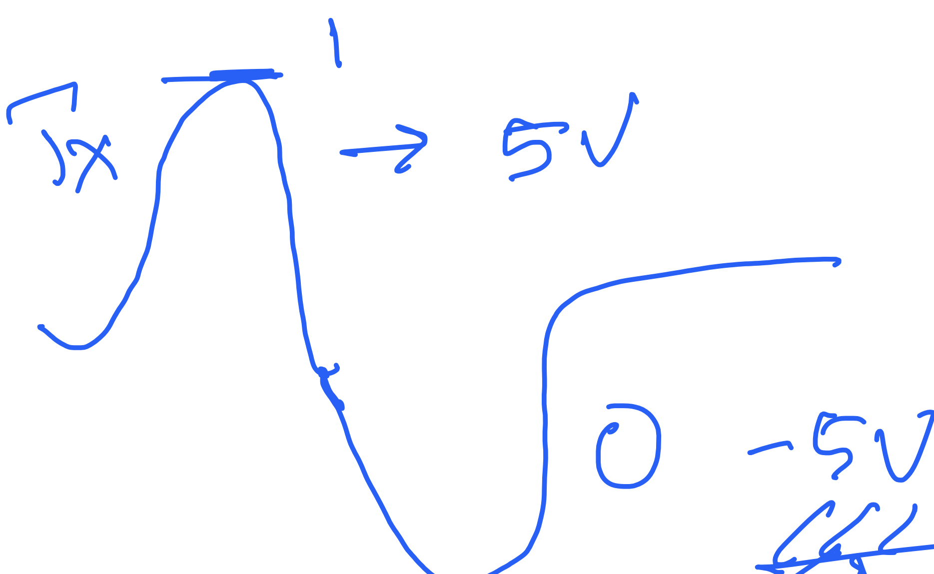




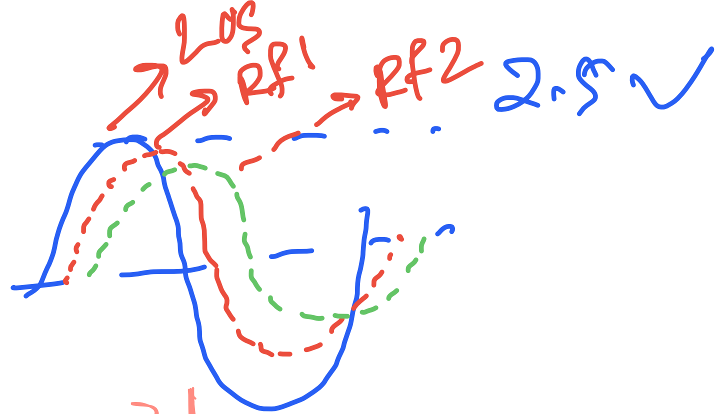
→ baseband



→ Carrier wave

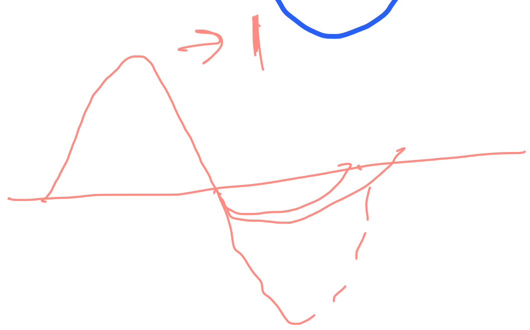


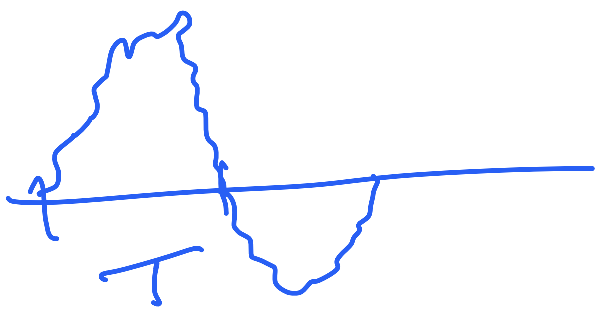
Rx



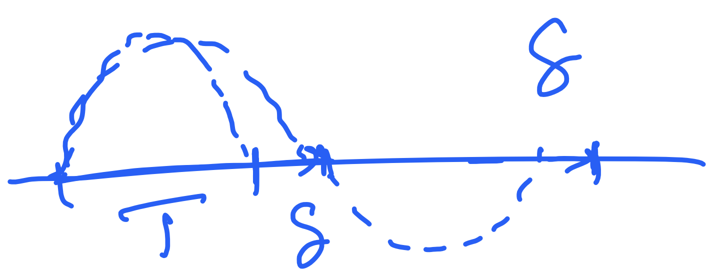
→ LOS

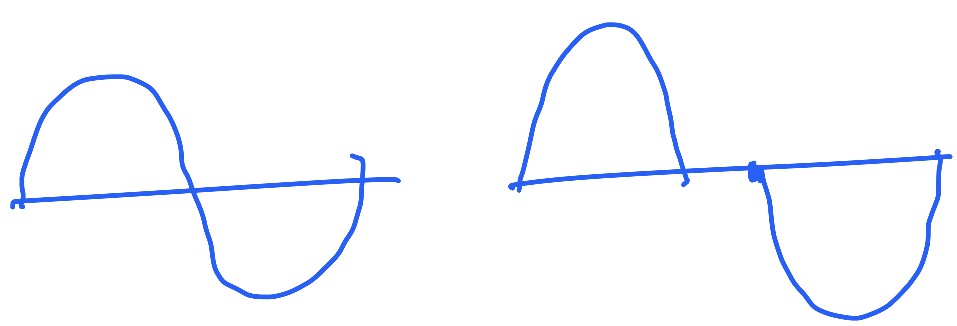
ISI → Inter Symbol Interference



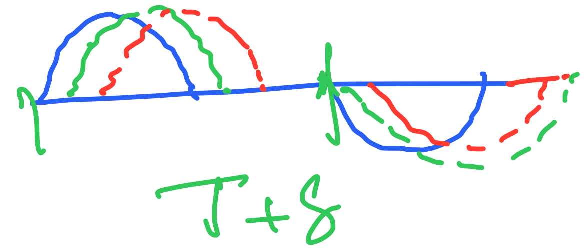


→ Receiver will actually receive.



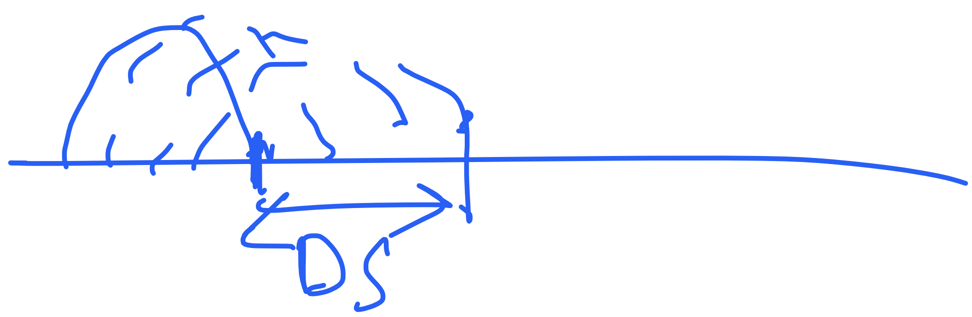


Tx



Delay spread → ~~time~~ signals travelling along diff path with diff length will arrive at the receiver at diff points of time

Delay spread represents time diff between when very first copy came & when the last came.



Guard band → time difference given between two symbols

Symbol duration → time for which you send a pulse + guard band